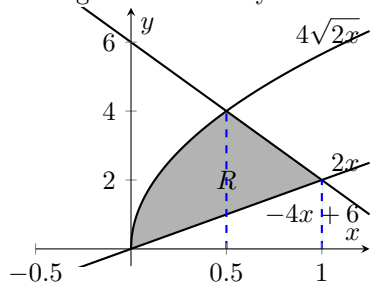


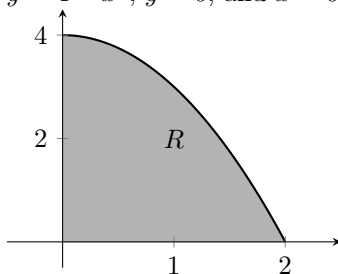
Justify your answers and show your work.*Find the area of the regions described in the problem.*

1. The region bounded by the functions $y = 4\sqrt{2x}$, $y = 2x$, and $y = -4x + 6$.

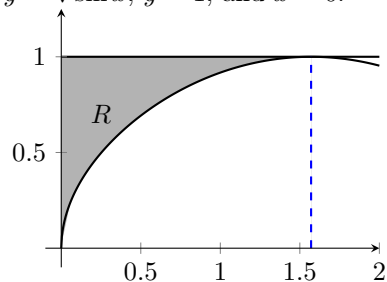


Let R be the region bounded by the following curves. Use the disk/washer method to find the volume of the solid generated when R is revolved about the x -axis.

2. $y = 4 - x^2$, $y = 0$, and $x = 0$.



3. $y = \sqrt{\sin x}$, $y = 1$, and $x = 0$.



Use the general slicing method to find the volume of the following solids.

4. The pyramid with a square base of 4 meters on a side and a height of 2 meters (use calculus).

5. A circular cylinder of radius of radius r and height of h whose axis is at an angle of $\pi/4$ to the base.